



Rx.AI Multimodal

Voice & Camera-Powered Patient Check-In

Powered by Google Cloud Vertex AI

From Irrelevant Check-in Forms to Fully Multimodal Adaptive Forms

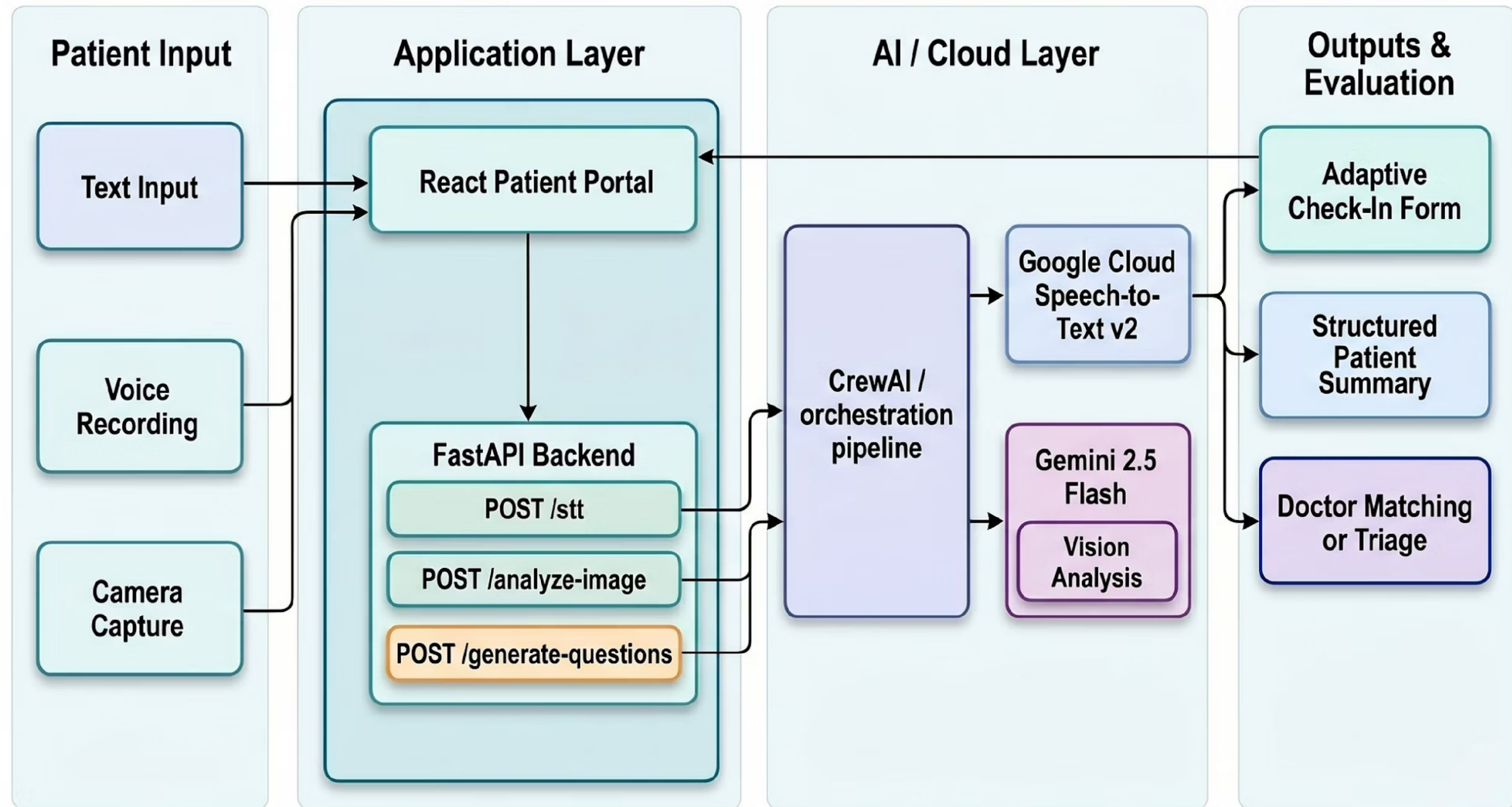
⚠ Problem

- Patient Check-in forms are fixed, often resulting in questions not related to the patient
- Patients with limited mobility or low digital literacy struggle to complete forms
- No way to collect wound photos, medication labels, or insurance cards inline

✅ Solution

- Dynamically generated questions based on patient history and patient's response to questions being asked
- Maintain patient summary, do preliminary triage and recommend relevant available doctors
- Voice output: Gemini TTS reads every question aloud
- Voice input: Google Cloud STT v2 transcribes spoken answers in real time
- Camera capture: Photo prompts triggered for relevant questions
- Graceful fallback: keyboard/click always available
- Rigorous evaluation: DeepEval metrics + BigQuery logging for every AI call

System Architecture



Evaluation Framework

STT (Speech-to-Text)	Word Error Rate, Medical Term Accuracy	100 audio clips with clinical transcripts
TTS (Text-to-Speech)	Naturalness & Intelligibility (LLM judge)	100 sample questions
Question Generation (CrewAI)	Relevance, Clinical Appropriateness, No Hallucination	200 patient profiles
Image Analysis	Description accuracy vs. human labels	300 labeled images (wound, medication, insurance card)

Workflow Combination Callouts



STT → Answer

Voice-transcribed vs. typed answer quality



Camera → Image description → Answer

Does visual context enrich responses?



Full Multimodal (STT + Camera)

Combined quality vs. text-only baseline